

Problem 3

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Problem 1 Estimate the value of $\sin\left(\frac{178\pi}{180}\right)$. Indicate whether your value is an overestimate or an underestimate.

Explanation. Set $f(x) = \sin(x)$. The problem is asking us to use linear approximation to estimate the value of $f\left(\frac{178\pi}{180}\right)$. Since $\frac{178\pi}{180}$ is very close to π , we will use L as the linearization of $\sin(x)$ at $a = \pi$.

$f'(x) = \cos(x)$ so $f'(a) = \cos(\pi) = -1$. $f(a) = \sin(\pi) = 0$. The linear approximation is given by $L(x) = -(x - \pi) + 0$.

$$\begin{aligned}\sin\left(\frac{178\pi}{180}\right) &\approx L\left(\frac{178\pi}{180}\right) \\ &= -\left(\frac{178\pi}{180} - \pi\right) \\ &= \frac{\pi}{90}.\end{aligned}$$

$f''(x) = -\sin(x)$. That means $f''(\pi) = 0$, BUT f'' is negative for all x in the interval $\left(\frac{178\pi}{180}, \pi\right)$. The graph of f is concave down across this interval, so the estimate is an overestimate.